

How colonization economies work

A player-facing guide for **Elite Dangerous** CMDRs using Raven Colonial to plan colonization ports, Odyssey settlements, hubs, and installations. It explains the same model as the [technical economy model](#), but without code terms.

What you see in Raven Colonial: every **Surface port**, **Orbital port**, and **settlement** with a market gets a percentage for each economy type: Agriculture, Refinery, Military, High Tech, Tourism, and so on. A value like **225%** means the market has economy strength **2.25** in the data.

Those percentages come from three places:

- The port's **host surface or orbital location**: planet type, moon type, star, asteroid cluster, signals, volcanism, rings, and similar conditions.
- The system's **reserve level**: Pristine, Major, Low, Depleted, etc.
- **Market links** from other completed builds in the system.

Spanish check: for completed in-game stations, Raven Colonial can compare its estimate with Spanish market snapshots. Planned and in-progress builds are for what-if planning only.

Surface and Orbital terms

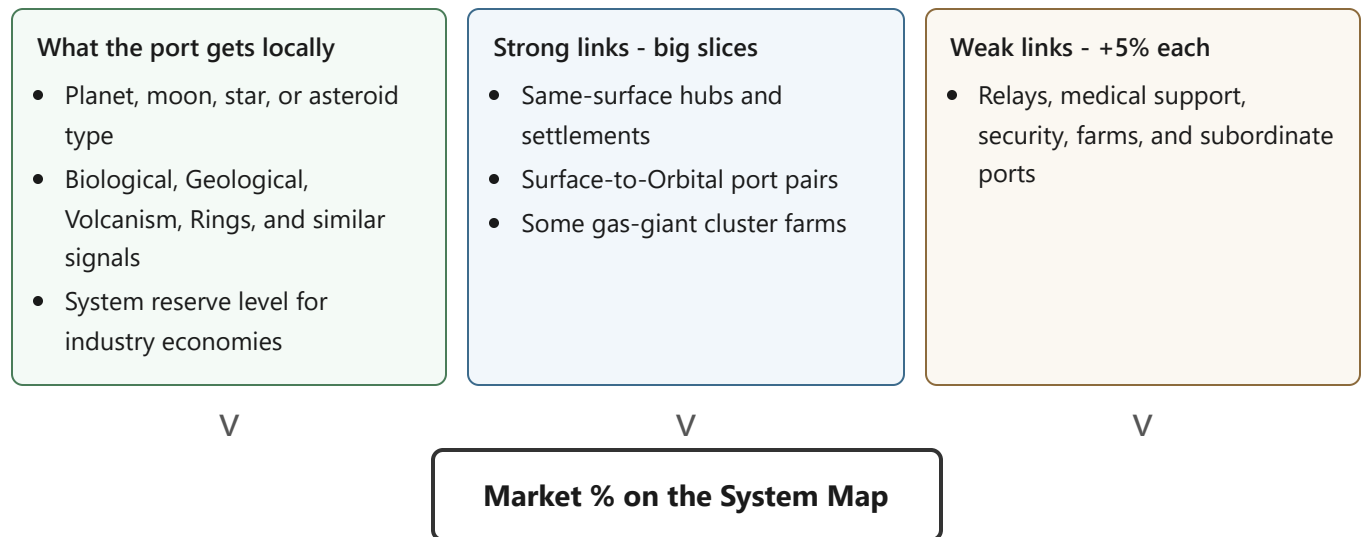
Raven Colonial uses these terms throughout this guide:

Term	Meaning
Surface port	A dockable starport or outpost built on the surface of a landable planet or moon
Orbital port	A dockable starport or outpost in space, including ports around a planet, moon, star, or asteroid cluster
Settlement	An Odyssey surface settlement
Hub / installation	A supporting build such as a relay, security post, space farm, comms installation, refinery hub, or science hub
Primary port	The main link receiver for a landable surface or orbital location. When both a Surface port and an Orbital port share the same planet or moon, the Orbital port is usually the primary
Subordinate port	An extra port attached under a primary port or hub

When this guide says **same surface**, it means builds on the same landable planet or moon. When it says **other surfaces**, it means a different planet or moon in the system.

The big picture

Think of each market row as a ledger. Raven Colonial adds every line that applies, then shows the total.



Mechanism	Typical amount	Do local conditions change it?
Local row	+100%, +50%, or +40% chunks	Yes
Strong link	+40%, +80%, or +120% per source	Yes, especially Agriculture and Pristine/Major industry
Weak link	+5% per source	No, weak links stay flat

Market Links shows candidates that can link into a port. For **Agriculture**, not every candidate is always used; the port has a limit on how many distant farm-style +5% links it can absorb.

Who counts: by default, only builds marked **Complete** in your colonization log affect the economy model. The **Include incomplete** beaker adds planned and in-progress builds for what-if planning.

What to build for each economy

This is the short CMDR version. The later sections explain the edge cases.

Goal	Typical helpers
Agriculture	Biological surfaces, Earth-like / Water worlds, space farms, Agriculture settlements
Refinery / Industrial / Extraction	Rocky, Rocky-Ice, Icy, HMC, Metal-rich, rings, Geological / Volcanism, refinery / industrial / extraction hubs
High Tech	Relays, medical installations, scientific settlements/hubs, Ammonia / Earth-like / Biological / Geological surfaces
Military	Security installations for weak links, military hubs for local strong links, star-based ports
Tourism	Tourist builds, Earth-like / Water / Ammonia surfaces, Biological / Geological signals, black holes, neutron stars, white dwarfs

Ports, settlements, hubs, and installations

Dockable ports are the main market rows most CMDRs care about. General colony ports use the full model: local row, strong links, weak links, then primary economy selection.

Surface ports and **Orbital ports** can behave differently when they share the same planet or moon. If a Surface colony port and an Orbital colony port are both present, the Surface port can push its local economies into the Orbital port as a strong link.

Specialized ports start with their specialty:

Specialized port location	Starting specialty
Surface	+50%
Orbital	+100%

Examples: Refinery outpost, Tourism starport, Extraction outpost.

Odyssey settlements have a shorter path. They get a fixed settlement economy, local buffs, and any settlement floors. Strong and weak links do **not** apply to the settlement's own market row, but settlements can still send links outward when the link graph makes them a source.

Hubs and installations mostly feed links into ports. If they have their own market economy, Raven Colonial gives them a fixed facility row rather than running the full port model.

How a colony port gets its numbers

1. Local body or orbital location

The port starts with economies from where it is built. Rocky surfaces add Refinery, Metal-rich and HMC surfaces add Extraction, Earth-like worlds add Agriculture / High Tech / Military / Tourism, stars add Military, black holes / neutron stars / white dwarfs add High Tech and Tourism, and so on.

2. Signals and reserve level

Biological and Geological signals add local rows where they apply. In **Pristine** or **Major** systems, Refinery, Industrial, and Extraction can gain **+40%**. In **Low** or **Depleted** systems, those same industry rows can lose **-40%**.

3. Strong links

Same-surface facilities, hub children, settlements, and Surface-to-Orbital port pairs can add large economy slices into a primary port.

4. Weak links

Eligible sources add **+5%** each. Weak links are intentionally small and do not get extra Pristine, Biological, Terraformable, tidal, or icy modifiers.

5. Primary economy

The highest total becomes the port's **primary economy**, which is the main economy tag the game highlights. Other non-zero economies still remain on the market row.

Strong links

Strong links are the main way to specialize a port. They are large because they represent local infrastructure directly feeding the port.

Strong link size

Strong link size follows **construction tier**, not pad size.

Source tier	Strong link contribution
T1	+40%
T2	+80%
T3	+120%

A T2 bio research settlement is a +80% strong-link source even if its pad is small.

Surface + Orbital pairs

If a Surface colony port and an Orbital colony port share the same planet or moon, the Surface port's local economies can strong-link into the Orbital port.

Agriculture has a special rule here: the Orbital port does not also repeat the same Biological / Earth-like / Water world agriculture buffs on its own row. Those bonuses move onto the strong link from the Surface port.

Hub children

A hub can bring its children along as **sub-strong** links. For example, a Refinery hub with settlements underneath can show the hub plus each child as strong-link contributors into the primary port.

In a **Pristine** or **Major** system, each Refinery / Industrial / Extraction strong-link contribution can also receive its own +40% reserve boost.

Gas-giant cluster farms

A space farm on a sibling moon under the same gas giant can strong-link Agriculture into that moon's **primary** port only. If a second port exists on that moon, it gets Agriculture through weak +5% steps instead of the farm's strong link.

Weak links

Weak links are background support from elsewhere in the system. Each applied weak link is +5%.

Who sends weak links

Sender	What it sends
Relay installation	High Tech +5% to ports on other surfaces or orbital locations
Medical installation	High Tech +5% support, and can help relays count for starports
Security installation	Military +5% to ports on other surfaces or orbital locations
Unanchored space farm	Agriculture +5% to other surfaces
Subordinate starport / outpost / hub	Its primary economy only, outward
Primary colony port	Agriculture outward only
Primary port at the central star	Agriculture outward only; Military and other local rows stay on that port
Military hub installation	No weak link; it strong-links Military locally instead

Relay and medical High Tech

Relays are straightforward on outposts: if the relay is eligible and it is not on the same surface as the receiver, it adds +5% **High Tech**.

Starports are pickier. A relay adds +5% **High Tech** to a starport only when the starport has a High Tech reason to receive it. That reason can be:

- High Tech already exists on the starport's market row, or
- Another non-relay High Tech weak source is linked, such as a medical installation or eligible science / high-tech support.

If a starport has no High Tech row and no non-relay High Tech support, the relay may still appear in **Market Links**, but it does not add +5% to the economy total.

Relays on the **same surface** as a port do not add the weak +5% High Tech to that port. They can still matter as local strong-link infrastructure where the build rules allow it.

Security posts

Security installations add +5% **Military** to ports on other surfaces or orbital locations. A security post on the same surface as your port does not add a Military weak link to that port.

Agriculture from distant stars

A Surface colony with no local Agriculture may accept only **one** Agriculture weak link per distant star's subtree. Farms and Agriculture ports around another star count once for that receiver. Agriculture settlements are exempt from that distant-star cap.

Agriculture

Agriculture is the special case because local agriculture, strong-link agriculture, and weak-link agriculture use different rules.

Observed multi-port setups often look like:

Port role	Common result
Subordinate Surface port on a Biological tidal moon	About 145%
Paired Orbital primary receiving the Surface port's strong link	About 225%

Local Agriculture row

On the port's own market row:

- Biological signals can add **+100% Agriculture** if the world type does not already grant Agriculture.
- Biological, Earth-like, and Water world conditions can add **+40%** Agriculture buffs.
- The optional **Terraformable Agri Bonuses** button can add a +40% what-if buff.
- Icy and tidal penalties do **not** reduce the port's own local Agriculture row.

Exception: when an Orbital colony port shares a planet or moon with a Surface colony port, the Orbital port skips those positive local Agriculture buffs. They arrive through the Surface-to-Orbital strong link instead.

Agriculture strong links

Strong-link Agriculture uses the receiver's conditions:

Receiver condition	Strong-link Agriculture effect
Biological	+40%
Earth-like / Water world	+40%
Terraformable toggle enabled	+40%
Icy / Rocky-Ice	-40%
Tidal to the star	-40%

A tidal or icy penalty cannot reduce an Agriculture strong link below **+10%**.

Agriculture weak links

Weak Agriculture is always **+5%** per applied farm, Agriculture port, or Agriculture settlement source. These +5% steps do not receive Biological, Terraformable, icy, tidal, or reserve modifiers.

Not every Agriculture source in **Market Links** is guaranteed to count. Raven Colonial limits how many weak Agriculture links a port can absorb. This limit is shown in the audit as skipped weak links when the cap is reached.

Common reasons for a tighter Agriculture weak-link limit:

- A same-surface Agriculture strong link already exists.
- The receiver is on Metal-rich, HMC, Icy, or Rocky-Ice terrain without being an Agriculture specialist.
- The receiver is an Orbital cluster-style port with several subordinates.
- The receiver is a habitable Agriculture-primary world with a small colony strong link.

A typical default cap is **90%** from weak links alone, which is eighteen +5% links, before tighter rules apply.

Terraformable Agri Bonuses

The System page has a **Terraformable Agri Bonuses** button. It is off by default because current Elite Dangerous market behavior does not appear to apply the expected Terraformable Agriculture modifier consistently.

Turn it on to preview what Agriculture numbers would look like if that in-game behavior is fixed later. The setting is saved in your browser.

Refinery, Industrial, and Extraction in Pristine systems

In **Pristine** or **Major** systems, reserve level matters in two places:

Where	Reserve effect
Port's local Refinery / Industrial / Extraction row	+40% once
Each Refinery / Industrial / Extraction strong link	+40% per strong-link contribution

In **Low** or **Depleted** systems, the same rows can receive **-40%** instead.

Example: Orbital starport over a rocky moon in a Pristine system.

Ledger line	Amount
Rocky surface gives Refinery	+100%
Pristine buff on local row	+40%
Strong link from Surface colony partner	+40%
Pristine buff on that strong link	+40%
Sub-strong link from Refinery hub child	+80%
Pristine buff on that sub-strong link	+40%
Total Refinery	340%

High Tech and Tourism body bonuses on strong links can apply at most once per port calculation. Pristine industry boosts are different: they stack per Refinery / Industrial / Extraction strong-link contribution.

Multiple ports on one surface

Frontier's colonization rules make multiple dockable ports around one planet or moon awkward. Raven Colonial models the behavior this way:

1. There is one **primary** port for that surface. If both a Surface port and an Orbital port are present, the Orbital port usually wins.
 2. Extra ports become **subordinates** and share the primary's weak-link candidates.
 3. A Surface colony port can strong-link its local economies into the paired Orbital colony port.
 4. Space farms strong-link the primary only. Other ports on the same surface use weak Agriculture +5% steps.
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Facilities you will build often

Facility	Effect on ports
Relay	High Tech +5% weak link to eligible ports away from the relay's own surface
Medical installation	High Tech +5% weak support; helps starports qualify for relay High Tech
Security post	Military +5% weak link to ports away from the security post's own surface
Space farm	Strong Agriculture into the local or gas-giant-cluster primary; weak Agriculture to other surfaces when unanchored
Refinery / Industrial / Extraction hub	Strong link of that economy into the primary port
Comms	Unlocks 140% High Tech on an Athena science hub when operational at the same location or on a parent planet, gas giant, or star
Military hub	Strong Military locally; does not weak-link outward

Documented rules and observed tuning

Most of the model follows community colonization references: body economy tables, +40% / +80% / +120% strong links by tier, +5% weak links, Surface-to-Orbital behavior, Agriculture's split rules, relay scope, and security scope.

Some parts are tuned against observed markets where public guidance is incomplete. The main tuned areas are Agriculture weak-link caps, a few settlement floors, and a small number of observed colony presets.

Known nuance: Low / Depleted industry penalties still apply on a port's local industry row today. Those may eventually move closer to Agriculture's behavior if in-game evidence changes.

Reading the economy table

Each audit line is one ledger entry:

- +40 Buff: body has BIO
- +80 Apply sub-strong link from: ...
- +5 Apply weak link from: ...
- Skipped weak link from: ... cap reached

Add the lines for one economy type to get the displayed percentage. For example, **145%** means **1.45** in the market data.

Primary economy is whichever economy type has the highest final total.

Scenarios worth knowing

Situation	What to expect
Gas-giant moons + ornamental starport	Relay may show in Market Links but not add High Tech unless the starport has a High Tech row or non-relay High Tech support; sibling-moon farms strong-link the primary only
Surface + Orbital colony pair	Surface port can strong-link local economies into the Orbital port; Orbital Agriculture buffs may move to the strong-link layer
Pristine rocky moon	Refinery can stack hard because Pristine applies to the local row and each Refinery strong link
Security on the same surface	Security does not add Military weak link locally; it supports ports on other surfaces or orbital locations
Port at the central star	Military can exist on the local row, but weak links sent outward are Agriculture only

Quick reference

Question	Answer
How big is a weak link?	Always +5%
How big is a strong link?	+40% / +80% / +120% by source construction tier
Does tide hurt my Surface port's own Agriculture row?	No on the local row; yes on Agriculture strong links into other ports
Does Pristine help Refinery more than once?	Yes : once on the local row and once on each Refinery strong link
Does a relay always add High Tech?	Outposts : usually yes if eligible and not same-surface. Starports : only when the starport has High Tech already or another non-relay High Tech support link
Do same-surface relays or security posts add weak links?	No . Same-surface weak-link processing only applies Agriculture
What does a primary colony port send outward?	Agriculture only
What does a subordinate port send outward?	Its primary economy only
Does a farm on another gas-giant moon strong-link every port?	No . It strong-links that moon's primary port; other ports use weak Agriculture

For modifier tables, exact caps, and code ownership, see [economy-model.md](#).